

MALMO AP, Sweden

WMO: 026360

Lat: 55.523N

Lon: 13.379E

Elev: 72

StdP: 100.46

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.9	-7.9	-13.0	1.2	-9.3	-10.2	1.6	-6.5	14.1	4.2	12.9	4.5	2.2	350	0.661

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	26.1	18.3	24.4	17.9	22.9	17.2	19.8	23.7	18.7	22.5	17.9	21.6	4.5	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	13.2	21.3	17.3	12.5	20.3	16.5	11.8	19.5	56.9	23.8	53.5	22.5	50.8	21.6	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.5	10.3	9.3	-15.2	28.6	4.4	1.9	-18.4	29.9	-21.0	31.0	-23.4	32.1	-26.7	33.4		
(5)			-15.5	21.7	4.3	0.9	-18.6	22.3	-21.2	22.9	-23.6	23.4	-26.8	24.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.1	0.1	0.3	2.3	6.8	10.9	14.2	16.9	16.7	13.2	8.9	4.8	1.7	(6)
(7)		DBStd	6.88	4.01	3.76	3.54	3.48	3.26	2.67	2.55	2.58	2.68	3.18	3.47	4.21	(7)
(8)		HDD10.0	1429	306	271	238	108	27	1	0	0	6	57	158	258	(8)
(9)		HDD18.3	3761	564	504	496	347	231	126	59	63	155	292	407	517	(9)
(10)		CDD10.0	744	0	0	0	11	55	128	215	209	101	24	1	0	(10)
(11)		CDD18.3	34	0	0	0	0	1	3	16	14	1	0	0	0	(11)
(12)		CDH23.3	276	0	0	0	1	4	36	146	88	1	0	0	0	(12)
(13)		CDH26.7	29	0	0	0	0	0	4	19	6	0	0	0	0	(13)
(14)	Wind	WSAvg	4.6	5.1	5.2	5.1	4.7	4.5	4.4	3.9	4.0	4.2	4.5	4.7	4.9	(14)
(15)	Precipitation	PrecAvg	706	54	44	41	40	51	62	69	79	67	64	70	65	(15)
(16)		PrecMax	900	116	104	83	70	118	130	208	232	155	134	138	128	(16)
(17)		PrecMin	542	0	8	10	10	15	1	0	19	20	31	16	30	(17)
(18)		PrecStd	95	30	19	21	18	22	30	50	54	37	27	28	25	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.8	8.9	13.1	20.9	23.7	26.7	28.9	27.4	23.0	17.2	12.7	10.0	(19)
(20)			MCWB	8.1	7.6	8.8	14.0	16.6	18.2	19.5	18.9	17.4	14.4	11.8	9.5	(20)
(21)		2%	DB	7.1	7.4	10.7	17.1	21.3	24.0	26.7	25.8	20.9	15.2	11.2	8.9	(21)
(22)			MCWB	6.5	6.6	7.4	11.1	15.0	17.5	18.8	18.2	16.4	13.4	10.3	8.3	(22)
(23)		5%	DB	6.1	6.4	8.7	14.9	19.4	21.8	24.9	24.0	19.1	14.2	10.2	7.9	(23)
(24)			MCWB	5.6	5.6	6.6	10.0	13.8	16.2	18.1	18.0	15.3	12.8	9.4	7.4	(24)
(25)		10%	DB	5.1	5.2	7.2	12.9	17.2	20.0	23.1	22.1	17.8	13.2	9.2	6.9	(25)
(26)			MCWB	4.6	4.5	5.7	9.1	12.7	15.2	17.4	17.2	14.6	11.9	8.5	6.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.0	8.0	9.7	14.5	17.7	20.2	21.2	21.0	18.4	15.1	12.1	9.7	(27)
(28)			MCDWB	8.4	8.3	12.2	19.8	22.2	24.6	26.3	24.9	21.8	16.2	12.5	9.9	(28)
(29)		2%	WB	6.6	6.8	7.9	12.0	16.0	18.1	20.0	19.7	16.8	14.0	10.7	8.4	(29)
(30)			MCDWB	6.9	7.3	9.5	16.0	19.9	22.4	24.4	23.0	19.5	15.0	11.2	8.7	(30)
(31)		5%	WB	5.6	5.7	6.9	10.7	14.7	16.8	19.0	18.7	15.9	13.1	9.6	7.2	(31)
(32)			MCDWB	5.9	6.2	8.1	14.0	18.2	20.5	23.2	21.9	18.1	14.2	10.0	7.6	(32)
(33)		10%	WB	4.6	4.6	5.8	9.4	13.2	15.7	18.1	17.9	15.2	12.3	8.6	6.2	(33)
(34)			MCDWB	5.0	5.1	7.0	12.3	16.5	19.1	21.9	21.3	17.2	13.3	9.0	6.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.1	4.4	6.2	8.5	9.3	9.0	9.2	8.8	7.5	5.8	4.3	4.2	(35)
(36)			MCDBR	3.8	4.7	8.4	11.6	12.3	12.3	13.1	11.6	9.3	6.3	4.0	4.0	(36)
(37)			MCWBR	3.8	4.1	6.0	7.1	7.2	6.8	6.5	5.8	5.6	4.8	3.8	4.2	(37)
(38)		5% WB	MCDBR	3.7	4.5	7.5	10.2	10.6	10.7	10.8	9.7	8.2	5.8	3.9	3.9	(38)
(39)			MCWBR	3.8	4.1	5.8	6.6	6.9	6.6	6.0	5.6	5.3	4.7	3.9	4.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.313	0.336	0.358	0.392	0.388	0.382	0.403	0.405	0.373	0.357	0.337	0.304	(40)	
(41)		taud	2.292	2.313	2.306	2.278	2.346	2.387	2.348	2.351	2.420	2.401	2.336	2.264	(41)	
(42)		Ebn at Noon	600	729	804	827	854	863	834	806	785	699	564	513	(42)	
(43)		Edh at Noon	55	77	98	116	114	111	114	107	87	71	53	45	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.45	1.06	2.44	4.20	5.40	5.75	5.34	4.30	3.05	1.50	0.58	0.30	(44)	
(45)		RadStd	0.09	0.17	0.27	0.51	0.54	0.44	0.65	0.31	0.29	0.22	0.10	0.03	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.45	N/A	N/A	N/A	N/A	N/A	N/A	-176	N/A	N/A
(47)	Regional (60 neighbors)	+0.41	N/A	N/A	N/A	N/A	N/A	N/A	-168	+62	N/A

Nomenclature: See separate page